

(*) Melnik-et al.-Dremel: A Decade of Interactive SQL Analysis at Web Scale-VLDB2020

Overview of the Main Idea

Dremel is Google's system for running fast, interactive SQL queries over massive datasets (with trillions of rows) stored in a distributed file system. The paper looks back over ten years and explains how Dremel combined key ideas disaggregated compute and storage, in-place (in situ) data analysis, and columnar storage for nested data to make real-time analytics at web scale possible. These ideas became the foundation for Google BigQuery and influenced many modern cloud analytics platforms.

Key Findings/Takeaways

By separating storage, memory, and processing, Dremel can use resources more efficiently, scale up quickly, and let many users run queries at the same time. Its columnar format for nested data, in-memory shuffle layer, and adaptive (runtime-changing) query plans together enable queries over petabytes to finish in seconds. Over time, Dremel also added support for joins, better SQL, and managed storage showing that cloud analytics systems must keep evolving to meet user needs.

Description of the System & Modifications

Dremel started as a static tree-based system with local disks, but evolved to a fully disaggregated, serverless architecture using distributed file systems (GFS, then Colossus) for storage, a persistent in-memory shuffle layer, centralized scheduling, flexible execution DAGs, and shared SQL libraries (ZetaSQL).

Workloads/Benchmarks Used

The paper describes real Google workloads at petabyte scale, including log analysis, dashboards, ETL pipelines, and ad hoc analytics, with processing speeds of over 100 billion records per second across many users.